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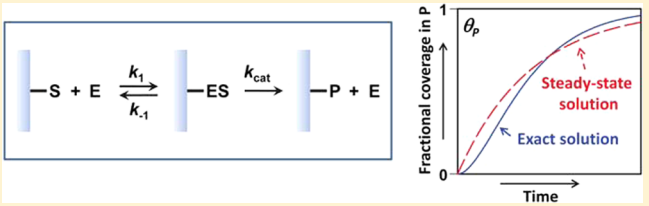
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Kinetics of Enzyme Action on Surface-Attached Substrates: A Practical Guide to Progress Curve Analysis in Any Kinetic Situation

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ABSTRACT: In the present work, exact kinetic equations describing the action of an enzyme in solution on a substrate attached to a surface have been derived in the framework of the Michaelis–Menten mechanism but without resorting to the often-used steady-state approximation. The here-derived kinetic equations are cast in a workable format, allowing us to introduce a simple and universal procedure for the quantitative analysis of enzyme surface kinetics that is valid for any kinetic situation. The results presented here should allow experimentalists studying the kinetics of enzyme action on immobilized substrates to analyze their data in a perfectly rigorous way.



INTRODUCTION

Whereas modern enzymology has reached a high degree of understanding of the catalytic action of enzymes on substrates in solution, much less is known about the behavior of enzymes toward substrates attached to a solid surface.[1,2] In the field of bioanalytical methods, a renewed interest in this unusual enzymatic configuration has been driven by the development of enzyme sensors using monolayers of substrate molecules immobilized on solid supports as sensing layers. The need to optimize the response of such sensors has motivated both theoretical[3,4] and experimental[5−14] research aimed at understanding the kinetics of enzymatic reactions at substrate-bearing interfaces. In early work, Okahata et al. characterized the kinetics of the cleavage reaction of surface-immobilized polysaccharide chains catalyzed by the glucoamylase and phosphorylase *b* enzymes.[5,6] At about the same period in time, the kinetic behavior of nuclease enzymes acting on monolayers of short DNA strands self-assembled on gold surfaces was analyzed by Corn et al.[8−10] The action of cutinase, a serine esterase enzyme, and of a kinase enzyme (Abl kinase) on their respective substrates attached to gold surfaces was similarly studied by Mrksich et al.[11−14] More recently, the kinetic behavior of the protease enzymes plasmin,[15] trypsin, and thrombin,[16,17] acting on dilute layers of redox-labeled peptide substrates immobilized onto gold electrode surfaces, was described by Takenaka et al.[15] and by our group.[16,17]

What is of interest here is that in most of the above-mentioned work the reaction of the enzymes acting on their attached substrates was kinetically modeled on the basis of the classical Michaelis–Menten equation, albeit this equation was derived for enzymatic reactions occurring in solution. In doing so, one implicitly assumed that the steady-state approximation, on which Michaelis–Menten equations are based, is also valid when the substrate is attached to a surface. However, Gutiérrez et al.[3] in their pioneering theoretical work and also Corn et al.[8,9] outlined that applying the steady-state-approximation to model the kinetics of enzyme action on immobilized substrates was problematic because in this heterogeneous configuration the condition of a large excess of substrate with respect to the enzyme is not met. As a way around this problem, Corn et al.[8−10] relied on numerical simulations to analyze their kinetic data and also proposed an empirically derived approximation as a replacement for the steady-state approximation. Paradoxically, exact kinetic equations for surface enzymatic reactions, derived without resorting to the steady-state approximation, can be found in the literature[3,4] but have never been used to analyze actual experimental data, probably because of the complex format in which they were presented.

In that context, the aims of the present article are (i) to establish on a mathematical basis the conditions under which the steady-state approximation is valid for modeling the kinetics of interfacial enzymatic reactions; (ii) to provide diagnostic criteria allowing experimentalists to identify situations where the steady-state approximation applies; and (iii) to derive exact analytical equations describing the kinetics of enzyme action on surface-attached substrates, accompanied by a detailed description of how to use these universally valid equations in order to analyze kinetic data rigorously in the most general situation.

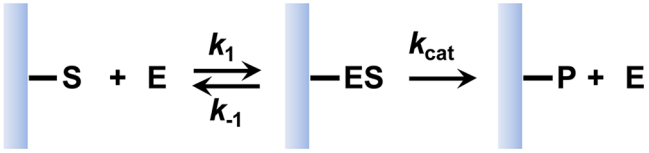
RESULTS AND DISCUSSION

In the framework of the Michaelis–Menten formalism, the action of an enzyme on a surface-attached substrate is represented by the following reaction sequence:

In the first, reversible, reaction, the enzyme, E, forms a precursor complex, ES, with the surface-attached substrate, S. In the second, so-called catalytic reaction, the complex

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dissociates irreversibly to release the enzyme in solution while leaving the product, P, on the surface. Because this system is heterogeneous, its kinetic behavior is potentially controlled not only by these two reaction steps but also by enzyme diffusion from the solution toward the surface. The kinetic contribution of this diffusional step can be taken into account mathematically but at the expense of very complex equations that can be solved only numerically.[4,8,9] Only in the specific case where no surface catalytic reaction occurs can analytical solutions be derived, but they then solely describe the binding kinetics.[18−20]

Fortunately, diffusional transport of the enzyme toward the electrode surface can experimentally be made fast as compared to the enzymatic reaction, rendering its contribution to the overall system kinetics negligible. This is typically achieved by mounting the substrate-bearing surface in a configuration allowing the rate of enzyme mass transport to be controlled and set at a sufficiently high rate as compared to that of the enzymatic reaction. Flow cells[8] or rotating disk setups[17,18] are typically used for this purpose. Hence, we restricted the theoretical analysis presented here to this most convenient and widely encountered fast enzyme diffusion limit. In this case, the enzyme concentration at the surface is constant and equals the bulk enzyme concentration, $[E]_t$.

As classically done for solution enzyme kinetics, the surface enzymatic action can be followed by monitoring the appearance of the product, P, as a function of time. In the present case, the surface concentration in the product, Γ_P, is measured and plotted as a function of time in a reaction progress curve. The analysis of the whole progress curve, or more often the analysis of its initial portion, yields global kinetic parameters whose dependence on the enzyme (and/or substrate) concentration(s) can then used to access the values of individual (or groups of) enzymatic rate constants.

However, such an analysis requires that a kinetic equation, relating the time-dependent surface concentration in the product to the rate constants of the Michaelis−Menten mechanism and to the initial concentration in the reactants, is established. Classically kinetic equations describing multistep enzymatic reactions occurring in solution are derived by making use of the widely known steady-state approximation. Within this approximation, the enzymatic reaction is schematically assumed to proceed in two successive phases: (i) a transient initial (so-called pre-steady-state) phase that is too fast to be resolved experimentally and during which a small amount of ES complex forms and (ii) a steady-state phase encompassing all of the observable experimental time, where the concentration in the enzyme−substrate complex does not vary with time because of the mutual compensation of the rates of complex formation, dissociation, and consumption by the catalytic reaction. It is also implicitly assumed that a negligible amount of substrate is consumed in the pre-steady-state phase. The above assumptions are easily met for enzymatic reactions in solution, where the substrate concentration typically greatly exceeds the enzyme concentration. As a result, the steady-state approximation is then legitimately applicable regardless of the actual relative values of the rate constants appearing in the mechanism. However, for the heterogeneous configuration considered here the amount of surface-attached substrate is in essence limited; therefore, a significant amount of the substrate can be consumed before a steady-state concentration in the enzyme−substrate complex is reached. This makes the use of the steady-state approximation for the kinetic modeling of enzyme action on immobilized substrates a priori questionable.[3,8] However, the applicability of the steady-state approximation for describing the kinetics of these heterogeneous enzymatic systems has never been theoretically assessed.

The only way to resolve this issue is to derive the exact general kinetic equation for such systems and compare the ensuing theoretical predictions to those derived in the framework of the steady-state approximation. This is precisely what is done below.

General Kinetic Equation for a Surface Enzymatic Reaction Following a Michaelis−Menten Mechanism. The set of differential equations describing the time variation of the surface concentrations Γ_i of the species i = S, ES, and P during the Michaelis−Menten surface reaction is

$$\begin{cases} \dfrac{\partial \Gamma_S}{\partial t} = k_{-1}\Gamma_{ES} - k_1[E]_t\Gamma_S \\ \dfrac{\partial \Gamma_{ES}}{\partial t} = k_1[E]_t\Gamma_S - (k_{-1} + k_{cat})\Gamma_{ES} \\ \dfrac{\partial \Gamma_P}{\partial t} = k_{cat}\Gamma_{ES} \end{cases} \quad (1)$$

The initial conditions ($t = 0$) are $\Gamma_S = \Gamma_0$ and $\Gamma_{ES} = \Gamma_P = 0$, where Γ_0 is the initial surface coverage in the substrate.

The conservation of the reaction "sites" on the electrode surface implies that

$$\Gamma_S + \Gamma_{ES} + \Gamma_P = \Gamma_0 \quad (2)$$

The above set of equations can be solved exactly to yield the theoretical Γ_P versus t progress curve.

However, to simplify the calculations, most of the kinetic studies of enzyme action on immobilized substrates make use of the steady-state approximation to derive the theoretical progress curve.

Derivations of both the steady-state and the exact solutions for the Γ_P versus t progress curve are detailed below.

Derivation of the Steady-State Solution for the Γ_P versus t Product Progress Curve. The kinetic steady-state approximation assumes that the intermediate species ES does not accumulate in the system (i.e., $\partial\Gamma_{ES}/\partial t = k_1[E]_t\Gamma_S - (k_{-1} + k_{cat})\Gamma_{ES} \approx 0$).

This equation allows Γ_{ES} to be related to Γ_S as $\Gamma_{ES} = k_1[E]_t\Gamma_S/(k_{-1} + k_{cat})$, which, once inserted into the conservation equation (eq 2), leads to $\Gamma_{ES} = (\Gamma_0 - \Gamma_P)/(1 + (K_M/[E]_t))$ with K_M being the classical Michaelis−Menten constant given by $K_M = (k_{cat} + k_{-1})/k_1$.

Hence the rate of production of P, as appearing in the above-given set of differential equations, becomes $\partial\Gamma_P/\partial t = k_{cat}(\Gamma_0 - \Gamma_P)/(1 + (K_M/[E]_t))$

The integration of this equation from $t = 0$ ($\Gamma_P = 0$) to any t and Γ_P values yields the final expression $\Gamma_P = \Gamma_0(1 - \exp[-k_{eff}t])$, with $k_{eff} = k_{cat}/\{1 + (K_M/[E]_t)\}$

It is also convenient to introduce $\theta_P^{ss} = \Gamma_P/\Gamma_0$, the fractional coverage in P as derived in the framework of the steady-state approximation, which is then given by

$$\theta_P^{ss} = 1 - \exp[-k_{eff}t] \quad (3)$$

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This latter simple monoexponential expression implies that the θ_P^{ss} versus t progress curve is a simple saturation curve, growing linearly with time at short times and monotonously reaching a limiting plateau value of 1 for $t \rightarrow +\infty$ (when the substrate conversion is complete).

As often underscored in the literature, eq 3 is similar to the one derived by Langmuir to describe irreversible adsorption kinetics,[21] and the corresponding progress curve is thus termed Langmuirian.

Experimental data progress curves have been very often analyzed on the basis of eq 3 by many authors.[5,6,11−14,16,17] In some cases, the Langmuirian equation (eq 3) is directly used to fit the whole experimental progress curve.[16,17] More often, the initial rate of product formation, V_0, is measured and analyzed on the basis of the corresponding theoretical expression derived from eq 3:[5,6,11−14]

$$V_0 = \left.\frac{\partial \theta_P^{ss}}{\partial t}\right|_{t=0} = k_{eff} = \frac{k_{cat}}{1 + (K_M/[E]_t)}$$

Derivation of the Exact Solution for the θ_p versus t Progress Curve. The set of eqs 1 can be solved without resorting to the steady-state approximation to yield exact solutions for the θ_P versus t progress curve but at the cost of more complex calculations, as follows.

By introducing Γ^*, the surface concentration of the unconverted substrate, defined as $\Gamma^* = \Gamma_S + \Gamma_{ES}$, we can simplify the set of eqs 1 to

$$\begin{cases} \dfrac{\partial \Gamma_{ES}}{\partial t} = k_1[E]_t\Gamma^* - (k_1[E]_t + k_{-1} + k_{cat})\Gamma_{ES} \\ \dfrac{\partial \Gamma^*}{\partial t} = -k_{cat}\Gamma_{ES} \end{cases}$$

The Laplace transformation converts this set of differential equations into the following set of simple linear equations (taking into account that for $t = 0$, $\Gamma_{ES} = 0$ and $\Gamma_S = \Gamma_0$)

$$\begin{cases} s\bar{\Gamma}_{ES} = k_1^*\bar{\Gamma}^* - \sigma\bar{\Gamma}_{ES} \\ s\bar{\Gamma}^* - \Gamma_0 = -k_{cat}\bar{\Gamma}_{ES} \end{cases}$$

where $\bar{\Gamma}_i$ denotes the Laplace transform of the surface concentration in species i, s is the Laplace variable, $k_1^* = k_1[E]_t$, and $\sigma = k_1^* + k_{-1} + k_{cat}$.

Hence

$$\bar{\Gamma}_{ES} = \frac{k_1^*\Gamma_0}{k_1^*k_{cat} + s(s + \sigma)}$$

and

$$\bar{\Gamma}^* = \frac{(s + \sigma)\Gamma_0}{k_1^*k_{cat} + s(s + \sigma)}$$

Upon performing an inverse Laplace transformation, one finally obtains the equations describing the variation of Γ_{ES} and Γ^* with time

$$\Gamma_{ES} = \frac{k_1^*\Gamma_0}{\sqrt{\Delta}}\left\{\exp\left[\left(\frac{\sigma - \sqrt{\Delta}}{2}\right)t\right] - \exp\left[-\left(\frac{\sigma + \sqrt{\Delta}}{2}\right)t\right]\right\} \quad (4)$$

$$\Gamma^* = \frac{\Gamma_0}{2\sqrt{\Delta}}\left\{(\sigma + \sqrt{\Delta})\exp\left[\left(\frac{\sigma - \sqrt{\Delta}}{2}\right)t\right] - (\sigma - \sqrt{\Delta})\exp\left[-\left(\frac{\sigma + \sqrt{\Delta}}{2}\right)t\right]\right\} \quad (5)$$

where $\Delta = \sigma^2 - 4k_1^*k_{cat} = (k_1^* - k_{cat})^2 + k_{-1}(2k_1^* + 2k_{cat} + k_{-1}) > 0$

The fractional coverage in P is then given by $\theta_P = 1 - (\Gamma^*/\Gamma_0)$.

Gutiérrez et al.[3] and Manimozhi et al.[4] have reported equations similar to eqs 4 and 5. Even though these two kinetic equations are valuable because they are exact analytical solutions, derived in the most general case, they appear to depend on too large a number of parameters to be used straightforwardly for data analysis. Hence, it is particularly relevant to derive a workable expression for the time dependence of θ_P.

To make eq 5 more tractable and to identify the minimum number of kinetic parameters actually required to describe the system, we introduce the following dimensionless variables $\theta^* = \Gamma^*/\Gamma_0$ and $\tau = \sigma(1 - 1/\lambda)t$ and we define a new parameter λ given by

$$\lambda = \frac{\sigma}{\sqrt{\Delta}} = \frac{1}{\sqrt{1 - (4k_1^*k_{cat}/\sigma^2)}} = \frac{1}{\sqrt{1 - \{4k_1^*k_{cat}/(k_1^* + k_{-1} + k_{cat})^2\}}} \geq 1$$

These changes lead to the following dimensionless formulation of eq 5:

$$\theta^* = \frac{1}{2}\left\{(1 + \lambda)\exp\left[-\frac{\tau}{2}\right] + (1 - \lambda)\exp\left[-\frac{1}{2}\left(\frac{\lambda + 1}{\lambda - 1}\right)\tau\right]\right\} \quad (6)$$

and to

$$\theta_P = 1 - \theta^* = 1 - \frac{1}{2}\left\{(1 + \lambda)\exp\left[-\frac{\tau}{2}\right] + (1 - \lambda)\exp\left[-\frac{1}{2}\left(\frac{\lambda + 1}{\lambda - 1}\right)\tau\right]\right\} \quad (7)$$

The major interest in eq 7 is that it captures all of the possible kinetic behaviors of the system through the value of a single parameter, namely, λ.

Figure 1 presents several θ_P versus τ progress curves calculated with eq 7 for several values of λ.

One can see that in the general case the variation of θ_P with τ (i.e., time) is an S-shaped (sigmoidal) saturation curve because of the biexponential nature of eq 7.

Two limiting situations can be identified, corresponding to extreme values of λ:

- $\lambda \rightarrow 1$. Equation 7 then becomes $\theta_P = 1 - \exp(-\tau/2)$.

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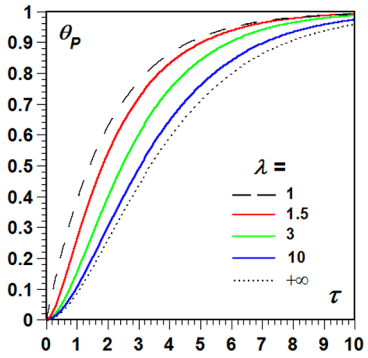


Figure 1. Theoretical progress curves showing the variation of the fractional coverage in the product, θ_P, as a function of the dimensionless time, τ, calculated using eq 7 for various values of the λ parameter.

This situation corresponds to the only case in which a monoexponential saturation curve is obtained (Figure 1, dashed line):

- $\lambda \gg 1$. Equation 7 then becomes

$$\theta_P = 1 - \left(1 + \frac{\tau}{2}\right)\exp\left[-\frac{\tau}{2}\right] \quad (8)$$

In this situation, the S shape of the progress curve is very marked (Figure 1, dotted line).

Hence, the value of λ decides the shape of the progress curve.

It is also worth noticing that the exact solution for the initial rate expression, V_0, in the general case is $V_0 = (\partial\theta_P/\partial\tau)_{\tau=0} = 0$, reflecting the fact that in the initial (pre-steady-state) phase of the reaction no product is actually formed. This result indicates that determining the kinetic parameters of the system from initial rate measurements may lead to very serious errors because a forced linear fit of the foot of the general case progress curve would yield an erroneous, artificially low apparent initial reaction rate V_0.

With the exact general solution for the progress curve in hand, we can now assess the conditions under which the steady-state approximation is valid.

Assessing the Validity of the Steady-State Approximation: Comparing the Exact and Steady-State Solutions for the θ_P versus τ Progress Curve. Within the new set of parameters, the steady-state solution for θ_P (i.e., θ_P^{ss}) can be written as

$$\theta_P^{ss} = \frac{\Gamma_P}{\Gamma_0} = 1 - \exp\left[-\left(1 + \frac{1}{\lambda}\right)\frac{\tau}{4}\right] \quad (9)$$

In Figure 2, representative steady-state progress curves calculated using this later expression are compared to the exact solutions for the θ_P versus τ variation derived from eq 7.

One can see from Figure 2 that in the general case (e.g., λ = 3) the Langmuirian progress curve derived from the steady-state approximation fails to describe the exact θ_P versus τ variation correctly: it does not reproduce the curved rising portion of the progress curve, it overestimates θ_P along most of the curve, and it underestimates θ_P at longer times. One also sees from Figure 2a that only when λ approaches unity can the progress curve be well described by the steady-state solution. Hence, the value of the λ parameter sets the applicability of the steady-state approximation as being fully valid only for the limiting situation $\lambda \to 1$.

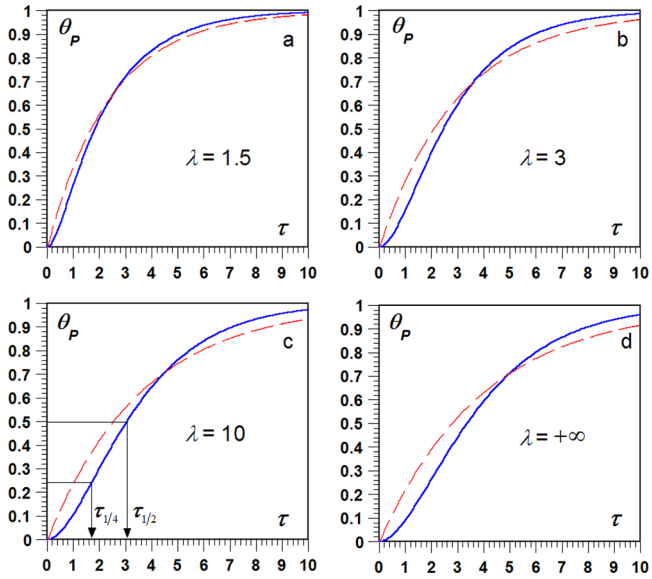


Figure 2. Comparison of the theoretical progress curves calculated using either the exact solution (eq 7, continuous blue line) or steady-state solution (eq 9, dashed red line). The progress curves were calculated for various values of λ, as indicated. The arrows in (c) show how the values of $\tau_{1/4}$ and $\tau_{1/2}$, the times for converting respectively one-quarter and one-half of the attached substrate molecules into product, can be derived graphically.

To estimate more quantitatively the error resulting from using the steady-state approximation in the general case, the value of $\tau_{1/2}$, the time required to convert half of the attached substrate molecules into product (i.e., the time required to reach θ_P = 0.5, see Figure 2c), is first calculated using eq 7. $\tau_{1/2}^{ss}$, the corresponding time for the steady-state solution, is then calculated using eq 9, and the $\tau_{1/2}/\tau_{1/2}^{ss}$ ratio is plotted in Figure 3.

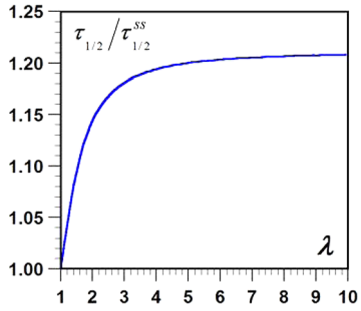


Figure 3. Quantitatively estimating the validity of using the steady-state approximation. Dependence of the $\tau_{1/2}/\tau_{1/2}^{ss}$ ratio on the value of λ. $\tau_{1/2}$ is the time required to convert half of the attached substrate molecules into product, calculated for the general case scenario (eq 7), and $\tau_{1/2}^{ss}$ is its counterpart calculated using the steady-state approximation (eq 9).

One can see from Figure 3 that $\tau_{1/2}^{ss}$ differs from $\tau_{1/2}$ by less than 10%, provided that λ < 1.5.

Kinetic Meaning of the Limiting Situations Identified. An important result of the present work is that we showed that the whole range of kinetic behavior of the system can be mathematically described on the basis of the value of a single dimensionless global parameter, λ. We now aim to establish the

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actual kinetic meaning of the limiting situations identified on the basis of this parameter.

We start by considering the $\lambda \rightarrow 1$ situation, which defines the validity of the steady-state approximation.

Recalling that λ is actually related to the rate constants of the reaction by the expression $\lambda = 1/[1 - \{4k^*_1 k_{cat}/(k^*_1 + k_{-1} + k_{cat})^2\}]^{1/2}$ we find that the limiting condition $\lambda \rightarrow 1$ actually corresponds to $4k^*_1 k/(k^*_1 + k_{-1} + k_{cat})^2 \ll 1$.

After some rearrangement, one can show that this condition is met, provided that

$$k_{-1}/k_{cat} \gg \Phi \tag{10}$$

where Φ is the following function of $[E]_t/K_M$:

$$\Phi = \frac{4[E]_t/K_M}{(1 + [E]_t/K_M)^2}$$

From eq 10 we see that the validity of the steady-state approximation actually depends on the ratio of the k_{-1} and k_{cat} rate constants but also on the enzyme concentration in solution through the $[E]_t/K_M$ ratio. More precisely, eq 10, and the steady-state approximation, are notably valid in the following limiting situations:

- $k_{-1}/k_{cat} \gg 1$ because Φ is always less than unity. In this particular case, the formation of the enzyme−substrate complex is at equilibrium, and the steady-state approximation is valid for any enzyme concentration in solution.
- $[E]_t/K_M \gg 1$ and also $[E]_t/K_M \ll 1$. The steady-state approximation is then valid independently of the value of the k_{-1}/k_{cat} ratio because for both of these limits $\Phi \rightarrow 0$. This later case has been previously identified by Gutiérrez et al.[3] but not quantitatively justified.

More generally, the above discussion suggests that the expression for λ is recast to be expressed as a function of kinetically relevant parameters k_{-1}/k_{cat} and $[E]_t/K_M$. In doing so, one finally obtains the following expression:

$$\lambda = \frac{1}{\sqrt{1 - \Phi/(1 + k_{-1}/k_{cat})}}$$

This latter equation shows that it is actually the full range of kinetic behaviors of the system that can be described on the basis of the k_{-1}/k_{cat} and $[E]_t/K_M$ parameters. This result allows us to construct the kinetic zone diagram presented in Figure 4, where the various limiting kinetic situations of the system are represented as portions of the $\log(k_{-1}/k_{cat}) - \log([E]_t/K_M)$ space.

The zones are respectively labeled LG for the Langmuirian case (i.e., $\lambda \rightarrow 1$), GC for the general case, and FS for the full sigmoidal case (i.e., $\lambda \gg 1$). The boundaries between the kinetic zones are defined as the (k_{-1}/k_{cat}, $[E]_t/K_M$) coordinates for which the $t_{1/2}$ value derived from the general case equation (eq 7) is within 10% of its counterpart calculated using the kinetic equation specific to each of the zones (i.e., eq 3 or 8).

As far as the LG ($\lambda \rightarrow 1$) limiting situation is concerned, an examination of the zone diagram yields the same general conclusions as those derived from eq 10 but yields more quantitative insights. In particular, Figure 4 reveals that a value of the k_{-1}/k_{cat} ratio greater than 1 is actually sufficient for the system to adopt Langmuirian-like behavior, no matter the enzyme concentration. One can also see that the system displays such behavior even for low values of the k_{-1}/k_{cat} ratio,

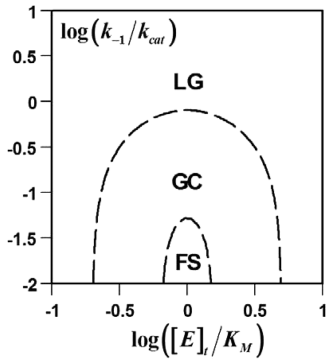


Figure 4. Zone diagram showing how the kinetic behavior of an enzyme acting on an immobilized substrate depends on the k_{-1}/k_{cat} ratio and on the enzyme concentration (normalized by K_M). The various limiting kinetic situations of the system are represented as portions of the $\log(k_{-1}/k_{cat}) - \log([E]_t/K_M)$ space and are defined as follows. LG, Langmuirian case ($\lambda \rightarrow 1$); GC, general case; and FS, full Sigmoidal case ($\lambda \gg 1$). The progress curves in cases LG, GC, and FS are respectively given by eqs 3, 7, and 8.

provided that $[E]_t/K_M \leq 0.2$ or $[E]_t/K_M \geq 5$. These conditions correspond to the cases where the rate-limiting step of the overall surface reaction is the (irreversible) formation of the enzyme−substrate complex or the catalytic reaction itself.

Quantitative insights into the physical meaning of the FS ($\lambda \gg 1$) limiting situation can also be gained from examining Figure 4: one sees that the FS zone is actually encountered only when the k_{-1}/k_{cat} ratio is low, in practice, lower than 0.05. Hence, this limiting situation corresponds to the case where the formation of the enzyme−substrate complex is irreversible and kinetically competes with the catalytic reaction to control the overall surface kinetics.

Experimental Diagnostic Criteria for Various Kinetic Situations: Practical Guide to Progress Curve Analysis in Any Kinetic Situation. One of the important consequences of the above theoretical predictions is that the shape of the progress curve can be used as a guide to identify the kinetic situation to which any given experimental system pertains. If the experimental progress curve rises uniformly toward a plateau following a monoexponential variation (i.e., it is not S-shaped), then the steady-state approximation can be used and is fully valid. In such a Langmuirian limiting situation, the values of k_{cat} and K_M can be experimentally determined by fitting the experimental progress curve with the Langmuirian eq 3 as classically done.[16,17]

However, if the progress curve is S-shaped (i.e., it presents a slowly rising "foot"), then the steady-state approximation does not hold. The system is then either in the general GC or the FS situation. To discriminate between these two cases experimentally, we propose that the ratio $t_{1/2}/t_{1/4}$, the times required to convert one-half and one-quarter of the attached substrate molecules to product, respectively, is measured on the progress curve and is used as a diagnostic criterion (Figure 2c). Indeed, as shown in Figure 5, the $t_{1/2}/t_{1/4}$ ratio solely depends on the λ parameter: for $\lambda \rightarrow 1$ (i.e., $\log(\lambda - 1) \rightarrow -\infty$), the value of the $t_{1/2}/t_{1/4}$ ratio tends toward 2.41, corresponding to Langmuirian behavior. For $\lambda \gg 1$ ($\log(\lambda - 1) \rightarrow +\infty$), the $t_{1/2}/t_{1/4}$ value tends toward 1.75.

Hence, an experimental $t_{1/2}/t_{1/4}$ ratio approaching 1.75 would indicate that the FS situation is met, whereas a $t_{1/2}/t_{1/4}$ value that is clearly above 1.75 would mean that the system is the general kinetic case.

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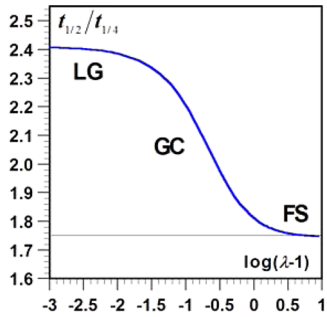


Figure 5. Working curve for experimentally determining the value of the λ parameter when the value of the $t_{1/2}/t_{1/4}$ ratio is known. $t_{1/2}$ and $t_{1/4}$ are the times required to convert one-half and one-quarter of the attached substrate molecules to product, respectively, and can be easily derived from progress curves (Figure 2c). The parts of the curve corresponding to kinetic situations LG, GC, and FS, as defined in the text, are indicated.

In the FS case, the experimental working curve has to be fitted using eq 8, yielding the value of the σ parameter, which in this limiting situation is simply given by $\sigma = 4k_{cat}/(1 + (K_M/[E]_t))$. Hence the k_{cat} and K_M values can then be derived from the enzyme concentration dependence of σ.

In the general case, a quantitative analysis of the progress curve is also possible but requires that the value of λ is experimentally determined from the experimental $t_{1/2}/t_{1/4}$ ratio using the working curve provided in Figure 5. The experimental progress curve can then be fitted using eq 7 with σ as the sole adjustable parameter. Determining the experimental values of the σ and λ parameters in such a way for various enzyme concentrations finally allows the rate constants appearing in the Michaelis–Menten surface reaction to be derived as follows. Recalling that $\sigma = k_1^* + k_{-1} + k_{cat} = k_1[E]_t + k_{-1} + k_{cat,}$ we see that the values of k_1 and $(k_{-1} + k_{cat})$ can be derived by simple linear regression of a σ versus $[E]_t$ plot. Besides, because $\lambda = 1/(1 - 4k_1^*k_{cat}/\sigma^2)^{1/2} = 1/(1 - 4k_1[E]_tk_{cat}/\sigma^2)^{1/2}$, knowing the λ value allows the k_1k_{cat} product to be determined. Hence, by combining the group of rate constants thus extracted, the individual values of all of the rate constants k_1, k_{-1}, and k_{cat} can be theoretically determined.

CONCLUSIONS

In the present work, exact kinetic equations describing the action of an enzyme in solution on a surface-attached substrate have been derived. It is shown that the kinetic behavior of the system can be fully described on the basis of a simple general master equation involving a single dimensionless parameter. Two limiting kinetic situations could be identified. The first one corresponds to the case where the often-used steady-state approximation is valid. We show that this situation is notably met when the formation of the enzyme–substrate complex is at equilibrium. The attainment of this limiting situation can be experimentally ascertained by the observation of a Langmuirian, monoexponential progress curve.

The second limiting situation corresponds to the case where the formation of the enzyme–substrate complex is irreversible and limits the overall kinetics jointly with the catalytic reaction. This situation is characterized by progress curves displaying a marked S shape.

Finally, between these two extreme cases (i.e., in the general situation), the progress curve is found to be non-Langmuirian and to display some degree of sigmoidicity.

Importantly, a simple universal procedure is proposed to analyze experimental progress curves in any kinetic situation. Experimental progress curves can then be fitted using the appropriate here-derived kinetic equations, yielding accurate values of (or groups of) kinetic constants controlling the system behavior. We hope that the present work will help experimentalists studying the kinetics of enzyme action on immobilized substrates to analyze their data in a perfectly rigorous way.

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Notes

The authors declare no competing financial interest.

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